

Which of the following procedures would mimic the actions of the parasympathetic division of the nervous system?

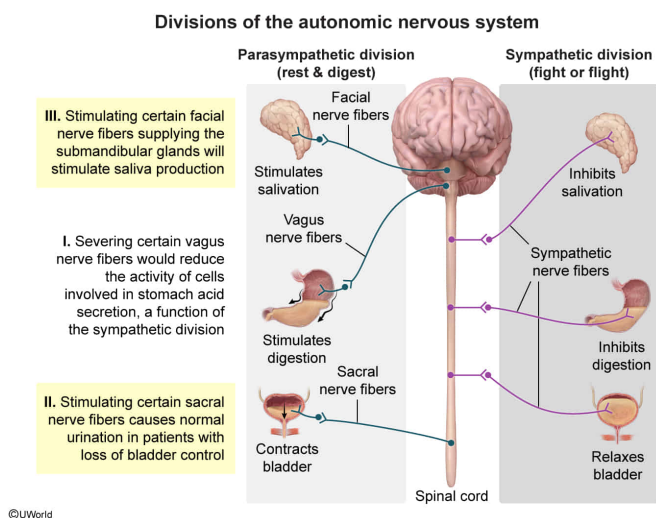
- I. Severing certain vagus nerve fibers to reduce the activity of cells involved in stomach acid secretion
- II. Stimulating certain sacral nerve fibers to cause normal urination patterns as a treatment for loss of bladder control
- III. Stimulating certain facial nerve fibers supplying the submandibular glands to improve saliva production

- ☐ A. II only [5%]
- ☐ B. III only [17%]
- ✓ ☒ C. II and III only [64%]
- ☐ D. I, II, and III [12%]

Correct

64% answered correctly

Explanation:



The **peripheral nervous system** is divided into:

- the sensory, or afferent, division, which is involved in transmitting impulses to the central nervous system (brain and spinal cord), and
- the motor, or efferent, division, which is involved in sending impulses to peripheral tissues and systems.

The motor division is subdivided into two branches:

- the somatic nervous system, which is responsible for activating voluntary skeletal **muscle**.
- the autonomic nervous system, which activates involuntary smooth muscle, cardiac muscle, and responses by glands (eg, **adrenal gland**).

Finally, the autonomic nervous system is subdivided into **two categories**:

- The sympathetic ("fight or flight") division is responsible for the mobilization of energy in stressful situations (ie, immediate danger). For example, stimulation of sympathetic division **nerve fibers** activates processes that lead to effects like increased heart rate and dilation of airways. At the same time, sympathetic activation inhibits activities such as digestion so that energy can be redirected toward addressing stressors.
- The parasympathetic ("rest and digest") division is responsible for energy storage when stress is low. For example, stimulation of the parasympathetic division promotes digestive activity and normal urination patterns.

Accordingly, the sympathetic and parasympathetic divisions are considered antagonistic because they promote effects that oppose one another.

This scenario requires identification of a procedure that would mimic the actions of the parasympathetic division. One function of the parasympathetic division is maintaining normal urination patterns when an immediate stressor is not present, while the sympathetic division suppresses urination during stressful situations. If certain sacral nerve fibers, which innervate the bladder and urinary tract, were **stimulated** to cause **normal urination patterns** as a treatment for loss of bladder control, this would mimic the actions of the **parasympathetic** division (**Number II**).

The parasympathetic division is also responsible for increasing saliva production, which increases digestive activity, while the sympathetic division suppresses digestion. The facial nerve functions in stimulating the production of saliva. If a patient underwent **stimulation** of certain facial nerve fibers supplying the submandibular glands to **improve saliva production**, digestive activity would be *increased*, one of the functions accomplished by the **parasympathetic** division (**Number III**).

(Number 1) The vagus nerve has a role in stimulating many body processes, including heart rate, the immune response, and digestion. By severing certain vagus nerve fibers to reduce the activity of cells involved in stomach acid secretion, digestive processes would be inhibited, a feature of the sympathetic division. The parasympathetic division would stimulate, *not* inhibit, digestion.

Educational objective:

The parasympathetic division promotes energy storage, while the sympathetic division promotes energy mobilization. The parasympathetic division and the sympathetic division have antagonistic effects.

Time spent:0 sec

Subject:

Biology

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System:

Endocrine and Nervous Systems